Technische Hochschule Ingolstadt

Fakultät Maschinenbau

Autonomes Fliegen

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Inhalt der Vorlesung 1/2

Kapitel

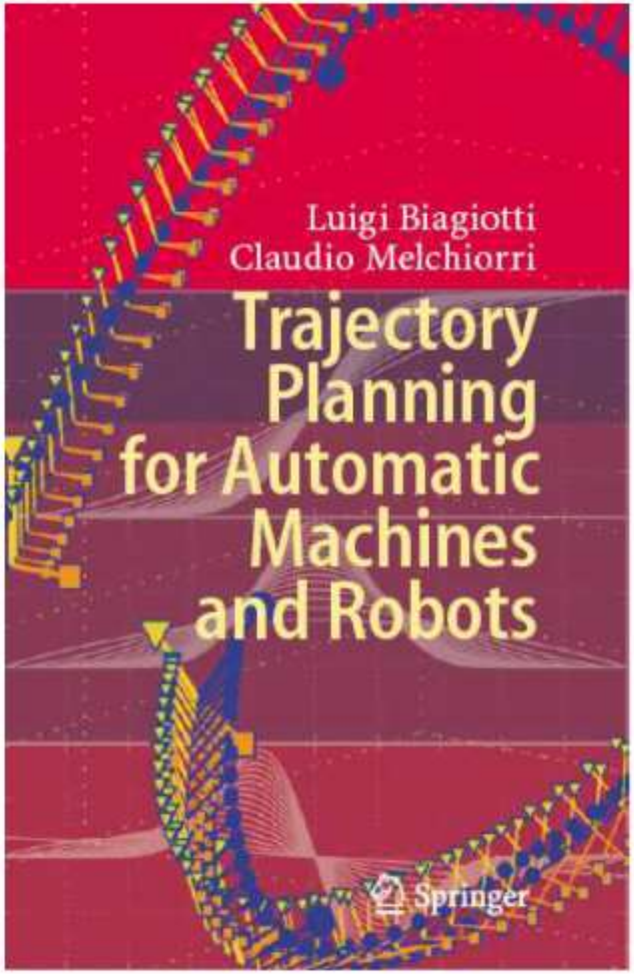
- **Kapitel 1: Übersicht über Unmanned Aerial Systems (UAVs) (1 W)**
- **Kapitel 2: Wesentliche Komponenten/Subsysteme eines UAV (1 W)**
- **Kapitel 3: Modellierung und Simulation von UAVs (1 W)**
- **Kapitel 4: Flight State Controller (PD Cascade) (1W)**
- **Kapitel 5: Pfadplanung (4 W)**
- **Kapitel 6: Perception und Sensing (1 W)**
- **Kapitel 7: Navigation (2 W)**
- **Kapitel 8: Multi-Vehicle Cooperation and Coordination (1 W)**

Trajectory Planning

Some suggested references:

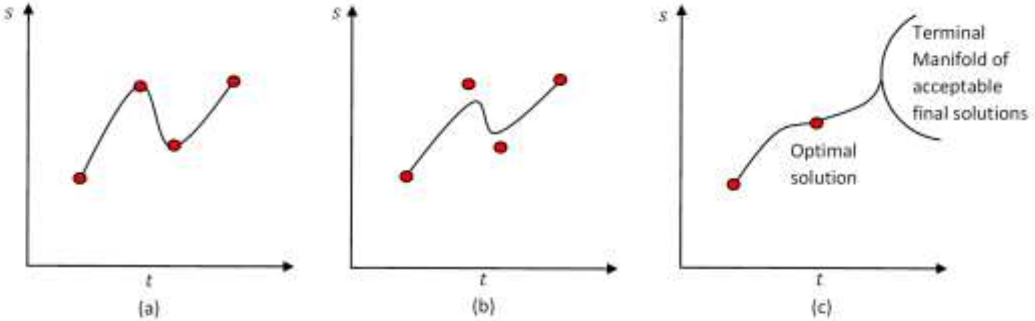
- C. Melchiorri, *Traiettorie per azionamenti elettrici*, Progetto Leonardo, Esculapio Ed., Bologna, Feb. 2000;
- G. Canini, C. Fantuzzi, *Controllo del moto per macchine automatiche*, Pitagora Ed., Bo, 2003;
- G. Legnani, M. Tiboni, R. Adamini, *Meccanica degli azionamenti: Vol. 1 - Azionamenti elettrici*, Progetto Leonardo, Esculapio Ed., Bologna, Feb. 2002.
- L. Biagiotti, C. Melchiorri, *Trajectory Planning for Automatic Machines and Robots*, Springer, 2008.

Trajectory Planning



Springer, 2008

- **Introduction**
- **Joint-space trajectories**
- **Polynomial trajectories**
- **Trapezoidal trajectories**
- **Spline trajectories**



Static interpolation (a), approximation (b), and dynamic interpolation (c)

Trajectory passing through a sequence of intermediate points

The results obtained with the polynomial (1) and the coefficients (3)-(6) can be generalized to the case in which $t_i \neq 0$. One obtains:

$$q(t) = a_0 + a_1(t - t_i) + a_2(t - t_i)^2 + a_3(t - t_i)^3 \qquad t_i \leq t \leq t_f$$

with coefficients

$$\begin{aligned}
a_0 &= q_i \\
a_1 &= \dot{q}_i \\
a_2 &= \frac{-3(q_i - q_f) - (2\dot{q}_i + \dot{q}_f)(t_f - t_i)}{(t_f - t_i)^2} \\
a_3 &= \frac{2(q_i - q_f) + (\dot{q}_i + \dot{q}_f)(t_f - t_i)}{(t_f - t_i)^3}
\end{aligned}$$

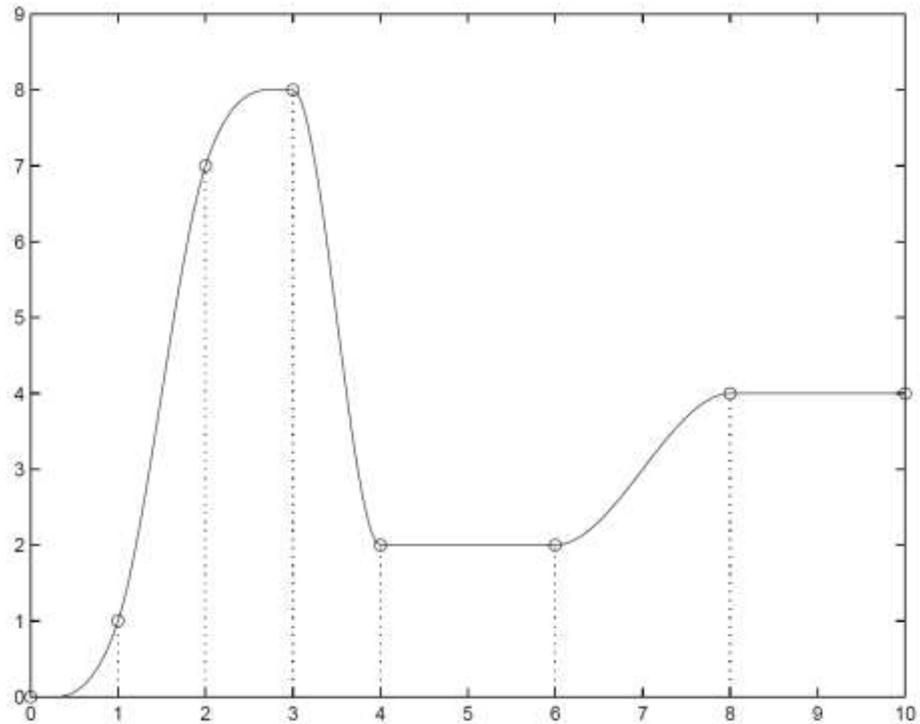
In this manner, it is very simple to plan a trajectory passing through a sequence of intermediate points.

Trajectory passing through a sequence of intermediate points

The trajectory is divided in n segments, each of them defined by:

- initial and final point q_k e q_{k+1}
- initial and final instant t_k, t_{k+1} $\qquad k = 0, \ldots, n-1.$
- initial and final velocity $\dot{q}_k, \dot{q}_{k+1}$

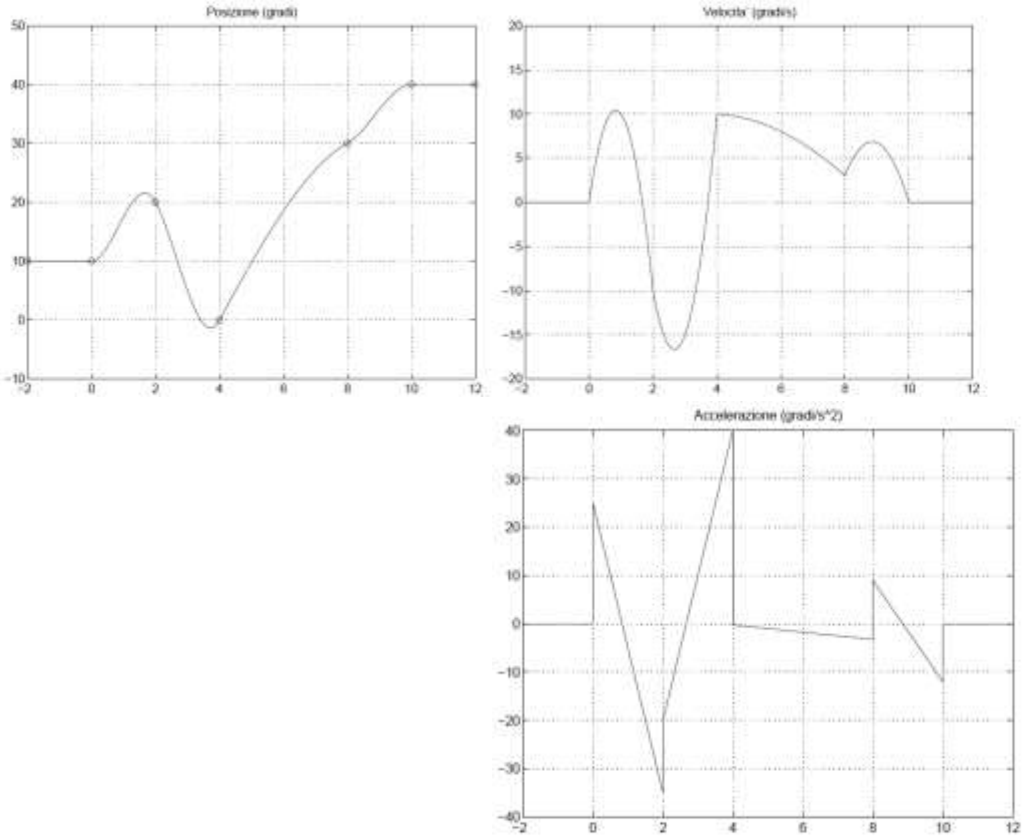
The above relationships are then adopted for each of these segments.



Trajectory passing through a sequence of intermediate points

Position, velocity and acceleration profiles with:

$t_0 = 0$	$t_1 = 2$	$t_2 = 4$	$t_3 = 8$	$t_4 = 10$
$q_0 = 10^o$	$q_1 = 20^o$	$q_2 = 0^o$	$q_3 = 30^o$	$q_4 = 40^o$
$\dot{q}_0 = 0^o/s$	$\dot{q}_1 = -10^o/s$	$\dot{q}_2 = 10^o/s$	$\dot{q}_3 = 3^o/s$	$\dot{q}_4 = 0^o/s$



Trajectory passing through a sequence of intermediate points

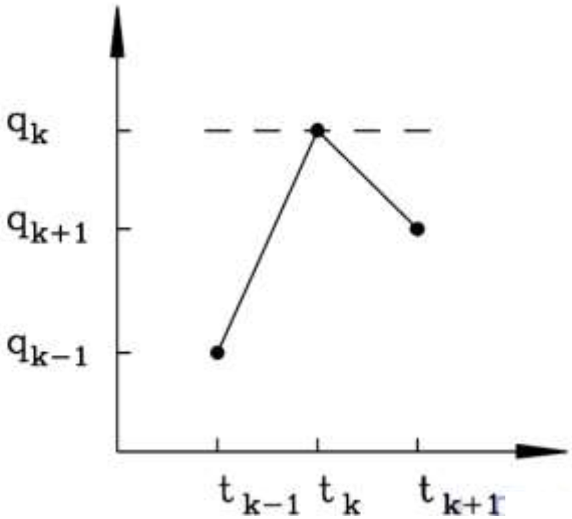
Often, a trajectory is assigned by specifying a sequence of desired points (*via-points*) without indication on the velocity in these points.
In these cases, the "most suitable" values for the velocities must be automatically computed.
This assignment is quite simple with heuristic rules such as:

$$\begin{aligned}
\dot{q}_1 &= 0; \\
\dot{q}_k &= \begin{cases} 0 & \text{sign}(v_k) \neq \text{sign}(v_{k+1}) \\ \frac{1}{2}(v_k + v_{k+1}) & \text{sign}(v_k) = \text{sign}(v_{k+1}) \end{cases} \\
\dot{q}_n &= 0
\end{aligned}$$

being

$$v_k = \frac{q_k - q_{k-1}}{t_k - t_{k-1}}$$

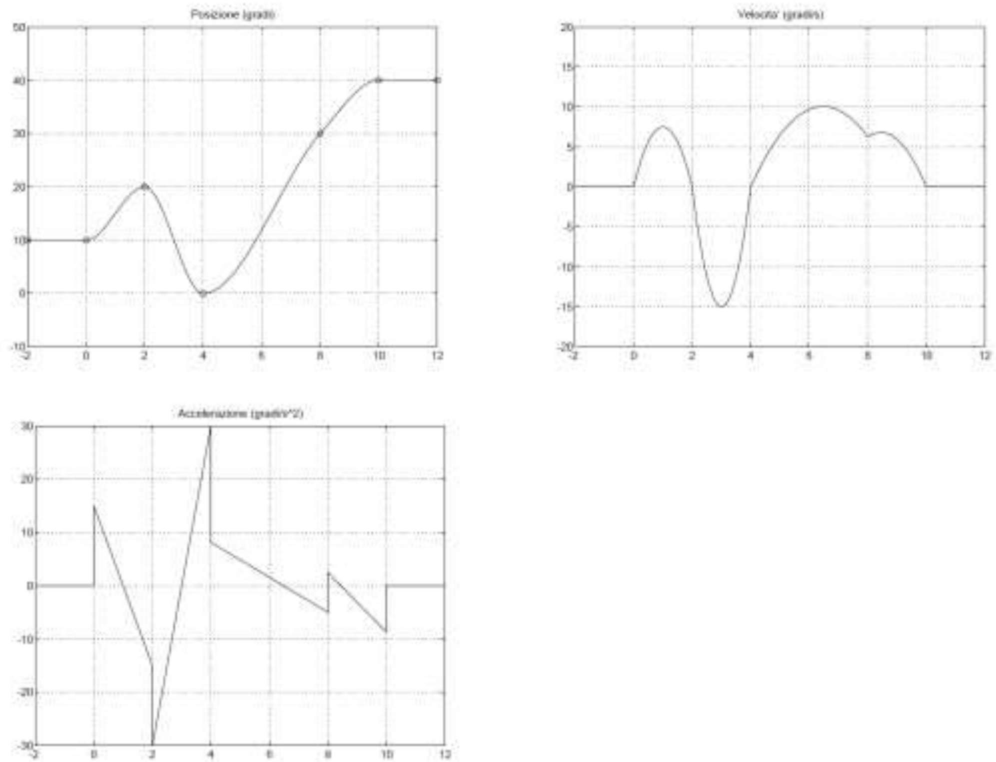
the 'slope' of the tract $[t_{k-1} - t_k]$.



Trajectory passing through a sequence of intermediate points

Automatic computation of the intermediate velocities (data as in the previous example)

$t_0 = 0$	$t_1 = 2$	$t_2 = 4$	$t_3 = 8$	$t_4 = 10$
$q_0 = 10^o$	$q_1 = 20^o$	$q_2 = 0^o$	$q_3 = 30^o$	$q_4 = 40^o$



Fifth-order polynomial trajectories

From the above examples, it may be noticed that both the position and velocity profiles are continuous functions of time.

This is not true for the acceleration, that presents discontinuities among different segments. Moreover, it is not possible to specify for this signal suitable initial/final values in each segment.

In many applications, these aspects do not constitute a problem, being the trajectories "smooth" enough.

On the other hand, if it is requested to specify initial and final values for the acceleration (e.g. for obtaining continuous acceleration profiles), then (at least) fifth-order polynomial functions should be considered

$$q(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$$

with the six boundary conditions:

$$\begin{array}{ll} q(t_i) = q_i & q(t_f) = q_f \\ \dot{q}(t_i) = \dot{q}_i & \dot{q}(t_f) = \dot{q}_f \\ \ddot{q}(t_i) = \ddot{q}_i & \ddot{q}(t_f) = \ddot{q}_f \end{array}$$

Fifth-order polynomial trajectories

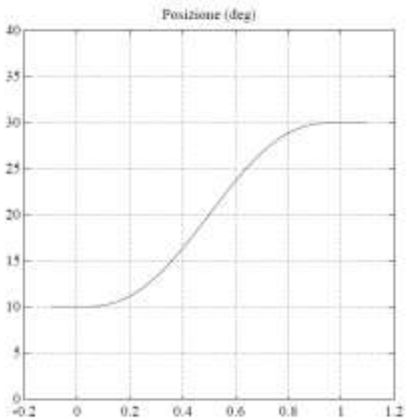
In this case (if $T = t_f - t_i$) the coefficients of the polynomial are

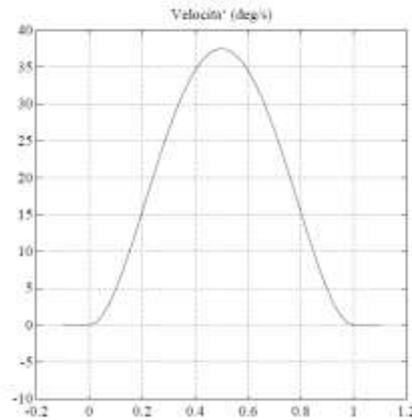
$$
\begin{aligned}
a_0 &= q_i \\
a_1 &= \dot{q}_i \\
a_2 &= \frac{1}{2}\ddot{q}_i \\
a_3 &= \frac{1}{2T^3}[20(q_f - q_i) - (8\dot{q}_f + 12\dot{q}_i)T - (3\ddot{q}_f - \ddot{q}_i)T^2] \\
a_4 &= \frac{1}{2T^4}[30(q_i - q_f) + (14\dot{q}_f + 16\dot{q}_i)T + (3\ddot{q}_f - 2\ddot{q}_i)T^2] \\
a_5 &= \frac{1}{2T^5}[12(q_f - q_i) - 6(\dot{q}_f + \dot{q}_i)T - (\ddot{q}_f - \ddot{q}_i)T^2]
\end{aligned}
$$

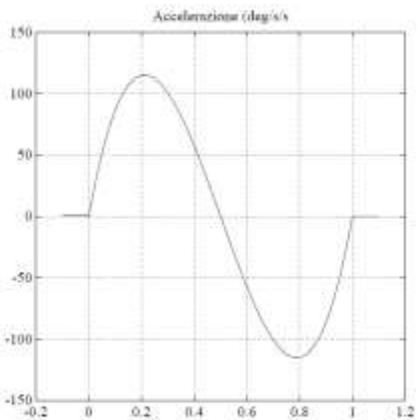
If a sequence of points is given, the same considerations made for third-order polynomials trajectories can be made in computing the intermediate velocity values.

Fifth-order polynomial trajectories

Fifth-order trajectory with the boundary conditions:
$q_i = 10^o, q_f = 30^o, \dot{q}_i = \dot{q}_f = 0\ ^o/s, \ddot{q}_i = \ddot{q}_f = 0\ ^o/s^2, t_i = 0s, t_f = 1s.$





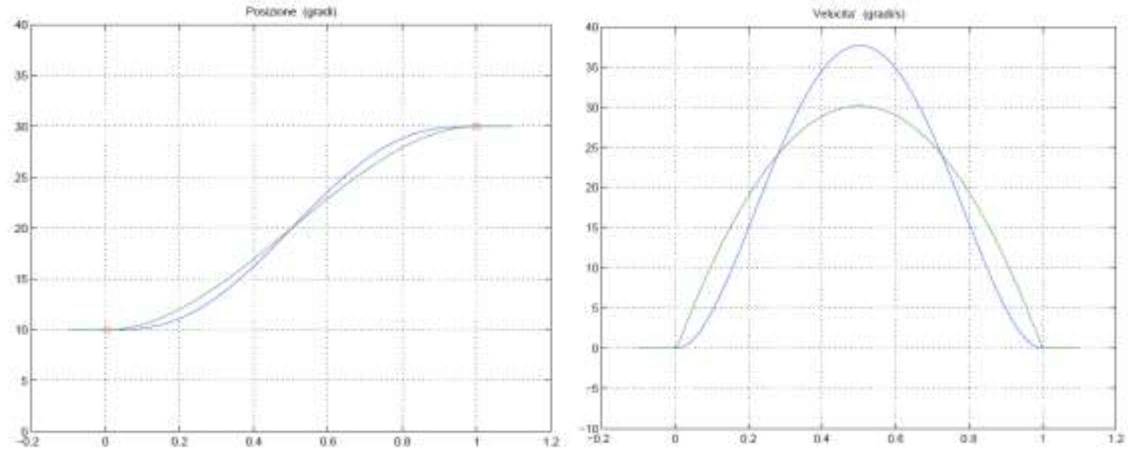


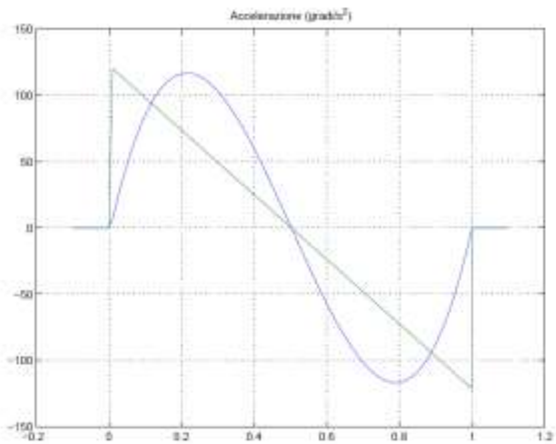
Obviously:

position	$\rightarrow$	5-th order function
velocity	$\rightarrow$	4-th order function
acceleration	$\rightarrow$	3-rd order function

Fifth-order polynomial trajectories

Comparison of fifth- and third-order trajectories with the boundary conditions:
$q_i = 10^o, q_f = 30^o, \dot{q}_i = \dot{q}_f = 0\ ^o/s, \ddot{q}_i = \ddot{q}_f = 0\ ^o/s^2, t_i = 0s, t_f = 1s.$

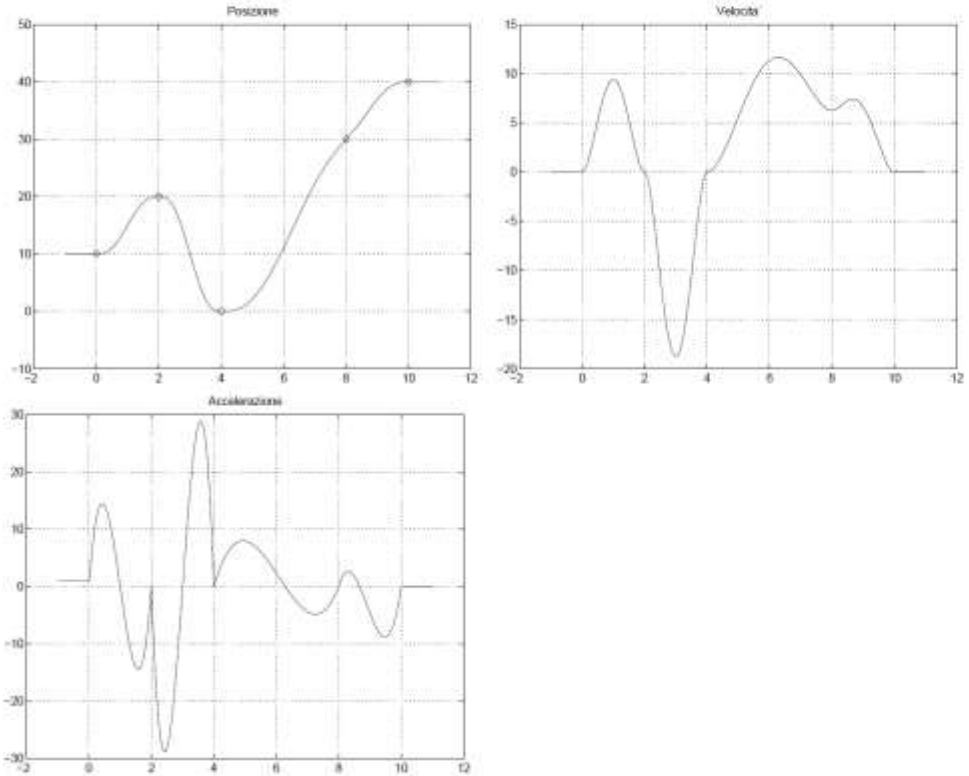




blue $\rightarrow$ fifth-order
green $\rightarrow$ third-order

Fifth-order polynomial trajectories

Position, velocity, acceleration profiles with automatic assignment of the intermediate velocities and null accelerations.



Note that the resulting motion has smoother profiles.